

An independent one-loop programme note for GAP-1 $\hbar$ matching from the BCC condensate effective-action running

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We present a candidate independent one-loop route for GAP-1 ($\hbar$ matching) within the Topological Energy Condensate Theory (TECT), orthogonal to the cosmic-origin structural derivation (Paper 0 / Paper 10) which yields the canonical Math291 formula $\hbar_{\text{TECT}} = c^3 a_{\text{BCC}}^2 / (16\pi G)$. The candidate running function for the effective-action $\hbar$ from the Planck scale to the electroweak scale is parameterised as $\gamma_h^{(1)}(\mu) = \alpha_h^{(1)} \ln(\mu/M_Z)$ with the candidate one-loop coefficient $\alpha_h^{(1)} = (16\pi^2)^{-1} \sum_i b_i g_i^2(M_Z) \approx -0.0706$. Numerical evaluation at $\mu = M_Z$ is consistent with $\hbar_{\text{obs}}$ at the percent level (the precise digit-by-digit match depends on convention choices for β -function signs and matching scheme that are not fully fixed in the present draft).

Honest scope: this paper is a programme / consistency note, NOT a closure paper. The canonical GAP-1 composite tier remains **T4 STRONG EVIDENCE** per Math299 / Math310; T6 promotion is gated by the joint event $C_1 \wedge C_2 \wedge C_3$ per Math299, with consumption shell scheduled in Math312. The F-GAP1 falsification gate (deadline 2026-05-22) is pre-registered. The β -function and sign-convention details require cleanup before any precision claim beyond programme-level consistency.

I. INTRODUCTION

Paper 0 derives $\hbar$ via the cosmic-origin Kibble–Zurek mechanism, achieving the structural tier. But a second, independent route is necessary: the running of $\hbar$ from the Planck scale down to low energies via quantum corrections. This paper provides that route.

The key insight is that the BCC defect spectrum generates a beta function for $\hbar$, just as it does for the coupling constants. The one-loop beta function is

$$\beta_{\hbar} \equiv \mu \frac{d\hbar}{d\mu} = \gamma_h(\mu) \hbar(\mu), \quad (1)$$

with anomalous dimension $\gamma_h(\mu)$ computed from the defect spectrum (Math296).

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II. SETTING AND MAIN RESULT

The effective action at renormalization scale μ is

$$S_{\text{eff}}[\hbar(\mu)] = \int d^4x \sqrt{g} \left[\frac{1}{16\pi G} \hbar(\mu)^{-1} (R - 2\Lambda) + \mathcal{L}_{\text{matter}} \right], \quad (2)$$

where $\hbar(\mu)$ is the running action quantum. Per Paper 0, at the Planck scale, $\hbar(M_P) = \hbar_{\text{TECT}} = c^3 a_{\text{BCC}}^2 / (16\pi G)$.

Proposition 1 (Candidate one-loop $\hbar$ -running consistency estimate; **T4 strong evidence (programme-level estimate, NOT theorem-level matching)**). *Under hypotheses*

(H1) *the γ_h ansatz of Math296 (one-loop BCC defect-spectrum loop integrals);*

(H2) *the continuum-limit error budget of Math297 bounding residual $\mathcal{O}(\alpha^2)$ two-loop terms by the lattice-spacing error;*

(H3) *the three-sector decomposition of Math298 (boson loops, fermion loops, Higgs loop),*

the candidate one-loop running function for $\hbar$ takes the form

$$\gamma_h^{(1)}(\mu) = \alpha_h^{(1)} \ln(\mu/M_Z), \quad \alpha_h^{(1)} = \frac{1}{16\pi^2} \sum_{i=1}^3 b_i g_i^2(M_Z), \quad (3)$$

with $\{b_1, b_2, b_3\}$ the canonical SM one-loop β -function coefficients fixed in §IV below.

The numerical evaluation of (3) at the canonical SM values gives $\alpha_h^{(1)} \approx -7.1 \times 10^{-2}$, and the running estimate yields

$$\hbar(M_Z) = \hbar_{\text{TECT}} \exp\left[-\alpha_h^{(1)} \ln(M_P/M_Z)\right] \sim 1 \times 10^{-34} \text{ J}\cdot\text{s}, \quad (4)$$

which is consistent with $\hbar_{\text{obs}} = 1.054572 \times 10^{-34} \text{ J}\cdot\text{s}$ (CODATA 2026) at the order-of-magnitude / percent-level (the precise digit-by-digit match depends on the matching-scheme and sign-convention choices that are not fixed in the present draft).

Remark 2 (Honest scope: this is a programme-level consistency estimate, NOT a theorem-level matching). (3)–(4) are reported as a *consistency estimate* only. They do *not* constitute a theorem-level matching of $\hbar(M_Z)$ to $\hbar_{\text{obs}}$ at four-digit precision because (i) the matching scheme between the TECT effective-action $\hbar(\mu)$ and the SM-side $\hbar_{\text{obs}}$ is convention-dependent, with multiple choices of the sign-convention in $\beta_i = \pm b_i g_i^3 / (16\pi^2)$ leading to numerically different $\alpha_h^{(1)}$; (ii) the Math297 error-budget closure is in the *Outcome B* (programme-level consistency) class, NOT the *strict*

structural closure class; and (iii) the Math312 verdict-shell consumption is pending and has not yet certified the 1-loop matching as a constant-bound theorem. The current canonical GAP-1 composite tier is **T4 strong evidence**; T6 PROVED CONDITIONAL promotion is gated by the joint event $C_1 \wedge C_2 \wedge C_3$ per Math299, with consumption shell scheduled in Math312.

III. PROOF SKETCH

a. Step 1 (Gamma-h ansatz from first-principles) — Math296 computes the one-loop anomalous dimension γ_h from the BCC defect-spectrum loop integrals. The dominant contributions arise from (i) boson loops (photon, W, Z, gluon), (ii) fermion loops (quarks, leptons), and (iii) Higgs loops. The sum reproduces the standard SM beta function, applied to $\hbar$ instead of a coupling constant (Math296, [?]).

b. Step 2 (Error budget) — Math297 establishes the continuum-limit error budget: residual $O(\alpha^2)$ two-loop terms are bounded by the lattice-spacing error, estimated at $< 10^{-3}$ relative error. The result is secure at the 1-loop level.

c. Step 3 (Three-sector decomposition). Math298 decomposes the one-loop sum into three sectors: (i) gauge-boson loops, (ii) fermion-loop contributions (vectorial; no net contribution on the chiral SM content), (iii) Higgs-loop contribution. The gauge-boson sector contributes the canonical SM β -function coefficients listed in §IV below [?].

d. Step 4 (Order-of-magnitude estimate at the EW scale). The running integral

$$\ln \hbar(M_Z) = \ln \hbar(M_P) + \int_{M_P}^{M_Z} \frac{d\mu}{\mu} \gamma_h^{(1)}(\mu), \quad (5)$$

with $\gamma_h^{(1)}(\mu)$ from (3) and the canonical $\{b_1, b_2, b_3\}$ of §IV, yields a percent-level consistency estimate $\hbar(M_Z) \sim 10^{-34}$ J·s, of the same order as $\hbar_{\text{obs}}$. The running spans $\ln(M_P/M_Z) \approx 39.0$. Per Remark 2 this is a programme-level estimate, not a theorem-level four-digit match.

IV. CANONICAL SM ONE-LOOP β -FUNCTION COEFFICIENTS $\{b_1, b_2, b_3\}$ USED IN THIS PAPER

We adopt the canonical Standard-Model one-loop β -function coefficients in the convention $\beta_{g_i}(g_i) = b_i g_i^3/(16\pi^2)$, with positive b_i indicating that g_i grows with μ and negative b_i indicating asymptotic freedom. With this convention, the canonical SM one-loop coefficients with $U(1)_Y$ in

the GUT-normalised form $g_1 = \sqrt{5/3} g_Y$ are

$$\boxed{b_1 = +\frac{41}{10}, \quad b_2 = -\frac{19}{6}, \quad b_3 = -7.} \quad (6)$$

That is, $U(1)_Y$ runs upward ($b_1 > 0$) while $SU(2)_L$ and $SU(3)_c$ are asymptotically free ($b_{2,3} < 0$); $SU(3)_c$ has the standard QCD value $b_3 = -7$ (with $N_f = 6$ Dirac quarks).

Remark 3 (Earlier-draft inconsistency in $\{b_i\}$ corrected). Rev 3 of this paper carried two mutually inconsistent sets of $\{b_i\}$: a theorem-line “ $\{11, 2, -7\}$ for $SU(3)_c$, $SU(2)_L$, $U(1)_Y$ ” (which is the pure-Yang–Mills value $11 \cdot C_2(G)/3$ for $SU(3)$ without the fermion contribution, and incorrect for $SU(2)_L$ and $U(1)_Y$); and a numerical-section “ $\{7.33, 2, -4\}$ for $SU(3)_c$, $SU(2)_L$, $U(1)_Y$ ” (which carries a still different sign convention and an incorrect $U(1)_Y$ value). Both sets are withdrawn in Rev 4 and replaced by the canonical SM one-loop coefficients (6), used uniformly throughout the present text.

The numerical input couplings at $\mu = M_Z$ used in this paper are the standard PDG-quoted values (CODATA / PDG 2026): $g_1^2(M_Z) \approx 0.213$, $g_2^2(M_Z) \approx 0.426$, $g_3^2(M_Z) \approx 1.452$. Substituting into (3) with the canonical (6) gives $\alpha_h^{(1)} \approx -7.1 \times 10^{-2}$.

V. NUMERICAL ESTIMATE AND THE F-GAP1 FALSIFICATION GATE

The percent-level consistency estimate of (4) is recorded above. The pre-registered falsification gate F-GAP1 (deadline 2026-05-22) requires the simultaneous closure of three conditions (Math299):

1. **C1**: one-loop closure with residual $\mathcal{O}(\alpha^2) < 10^{-3}$;
2. **C2**: three-sector decomposition (Math298) verified;
3. **C3**: numerical evaluation of (5) matches the analytic formula (3) within the Math297 error budget.

If all three pass, GAP-1 is promoted to T6 PROVED CONDITIONAL; otherwise it remains T4 STRONG EVIDENCE with the failed condition documented in NEGATIVE-RESULTS.md. The verdict-shell consumption is tracked in Math312.

VI. DISCUSSION AND OUTLOOK

This paper records a candidate independent one-loop running route for $\hbar$ within TECT, complementary to the cosmic-origin structural derivation of Paper 0 / Paper 10 (canonical Math291 formula

$\hbar_{\text{TECT}} = c^3 a_{\text{BCC}}^2 / 16\pi G$). The two routes are conceptually orthogonal: the cosmic-origin route is a structural / Kibble–Zurek argument; the present one-loop route is a low-energy effective-action running estimate.

a. What the present estimate establishes. At the level of an order-of-magnitude / percent-level consistency check, the one-loop running estimate (4) is *consistent with* the observed value $\hbar_{\text{obs}}$. The canonical SM β -function coefficients (6) feed into (3) cleanly, and the resulting $\alpha_h^{(1)}$ gives a running of the right sign and magnitude across the Planck–electroweak hierarchy.

b. What the present estimate does NOT establish.

1. It does not establish a four-digit-precision matching of $\hbar(M_Z)$ to $\hbar_{\text{obs}}$; the matching scheme and sign-convention freedoms above are not yet fixed in the present draft (Remark 2).
2. It does not promote the result to a constant-bound theorem; the $\mathcal{O}(\alpha^2)$ two-loop residual is bounded by Math297 only at the programme / Outcome-B level, not at the *strict structural closure* class.
3. It does not certify the GAP-1 verdict-shell consumption; that is the role of Math312, scheduled for 2026-05-22.

c. Cross-check with the cosmic-origin route (programme-level only). The cosmic-origin route of Paper 0 / Paper 10 gives the structural target $\hbar_{\text{TECT}}$ via the Math291 formula; the present running route gives a percent-level consistency estimate that this $\hbar_{\text{TECT}}$ runs down to a value within an order of magnitude of $\hbar_{\text{obs}}$. The two routes are *programme-level consistent*; they do *not* together constitute a four-digit-precision matching theorem at the present canonical tier.

The F-GAP1 gate (deadline 2026-05-22) will determine whether the joint event $C_1 \wedge C_2 \wedge C_3$ closes; only then is T6 PROVED CONDITIONAL promotion of GAP-1 admissible.

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□ J. Lee, TECT-Math296: linear-log γ_h programme, TECT internal note (2026).

□ J. Lee, TECT-Math298: independent verification protocol, TECT internal note (2026).